

A PROJECT REPORT ON

STUDY OF NON-THERMAL EMISSION FROM BLAZARS

Submitted by

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For the award of Degree of

BACHELOR OF SCIENCE (PHYSICS)
UNDER
UNIVERSITY OF CALICUT



**PG & RESEARCH DEPARTMENT OF PHYSICS
FAROOK COLLEGE (AUTONOMOUS)
FAROOK COLLEGE (PO), CALICUT**

APRIL 2023

CERTIFICATE

*This is to certify that project entitled “**Study of Non Thermal Emission from Blazars**” is done by Fathima Shifa M, Sixth semester physics students of Farook College under the guidance of Dr. Sulfikkarali N. K and Mrs Amal Abdul Rahman for the partial fulfilment of under graduate programme during the year 2020-2023.*

Examiner

Head of the Department

CERTIFICATE OF GUIDE

*This is to certify that project entitled “**Study of Non Thermal Emission from Blazars**” is done by Fathima Shifa M, Sixth semester physics students of Farook College under my guidance, for the partial fulfilment of under graduate programme during the year 2020-2023.*

Farook College

Date:

Dr Sulfikkarali N. k
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DECLARATION

*I hereby declare that this project entitled “**Study of Non Thermal Emission from Blazars**” submitted to Farook College (Autonomous), affiliated to the University of Calicut in partial fulfilment of the requirement for the award of Bachelor of Science is a bonafide record of the work carried by us during the period 2020-2023 under the supervision and guidance of Dr Sulfikkarali N. K, Assistant professor and Mrs Amal Abdul Rahman, Senior Research Fellow, Department of Physics, Farook College.*

Date: April 12,2023

Place: Farook College

Fathima Shifa M (FKAUSPH045)

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ABSTRACT

The spectrum of blazars are non-thermal in nature and extends from radio to γ -rays. Blazar's Spectral Energy Distribution is a double hump curve. The lower energy hump as understood originates due to synchrotron emission process. This extends from radio to x-rays. And the higher energy hump is understood to be due to Inverse Compton mechanism. The electron distribution responsible for synchrotron emission is found out to follow a broken power law trend. An analytical expression for synchrotron emissivity cannot be attained. In this project, we solve for an approximate analytical solution to synchrotron emissivity. The approximate analytical solution is used to understand the relationship between physical parameters and Spectral energy distribution of blazars. The theoretical model is applied on the SED of blazar Mkn 421 and estimated/constrained the physical parameters of the source..

Chapter 1

INTRODUCTION TO ACTIVE GALACTIC NUCLEI

1.1 Active Galactic Nuclei

An active galactic nuclei (AGN) is a compact region at the centre of some galaxy that has a much higher luminosity than normal galaxies. AGN are powered by supermassive black holes that accrete dust and cloud particles from surroundings. This accretion property of a black hole gives the barred spiral structures to our galaxy. They emit intense radiation across the electromagnetic spectrum and can have a significant impact on their host galaxies and surrounding environment. The radiation produced by them can be used to study the properties of black holes, the evolution of galaxies and the evolution of the universe as a whole. The galaxies with AGN in their centre are called active galaxies, in which their centres are more luminous than entire galaxies. A normal galaxy like our own Milky Way galaxy has a total luminosity of 10^{44} erg/s whereas just the central luminosity of an AGN exceeds 10^{44} erg/sec. AGN are classified based on their spectra and observed characteristics into different types, including Seyfert galaxies, quasars and blazars.

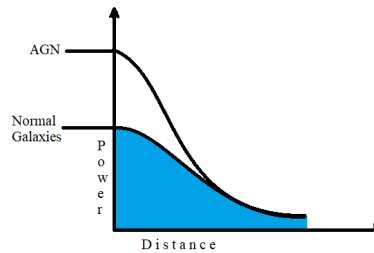


Figure 1.1: Central Luminosity of Normal galaxies and AGNs

History Of AGN:

In 1908, Dr Edward Fath from university of California at Lick Obesrvatory a series of observation to clarifying the nature of spiral nebulae,while working on NGC 1068 known by “Spiral Nebula” then observed an unusual line in some galaxy nuclei. Dr. vesto silpher later confirmed the reason behind that is due to redshift of spectrum and in 1943,Carl Seyfert published a paper in which he described the observations of some galaxies having bright nuclei that where source of broad emission lines, which later was confirmed due to Doppler broadening. During world war 2, radio engineers encountered some strange noise while using communication devices. The source of the noise was later determined to be celestial bodies like sun and Milky Way galaxy. This led to advancements in radio astronomy and the discovery of first radio galaxies, including Cygnus A, the first AGN found through radio observation. This motivated them to a question, if sky is also luminated in radio apart from optical. Why wouldn’t it be also luminated in other electromagnetic frequencies Active galaxies emit broader energy distribution. The emissions are in broader frequency range from Radio to X-Rays.

Active Galactic Nuclei or AGNs are a type of galaxies that emit with broader energy distribution. The emissions are in broader frequency range from Radio to X-Ray.

1.1.1 AGN Classification

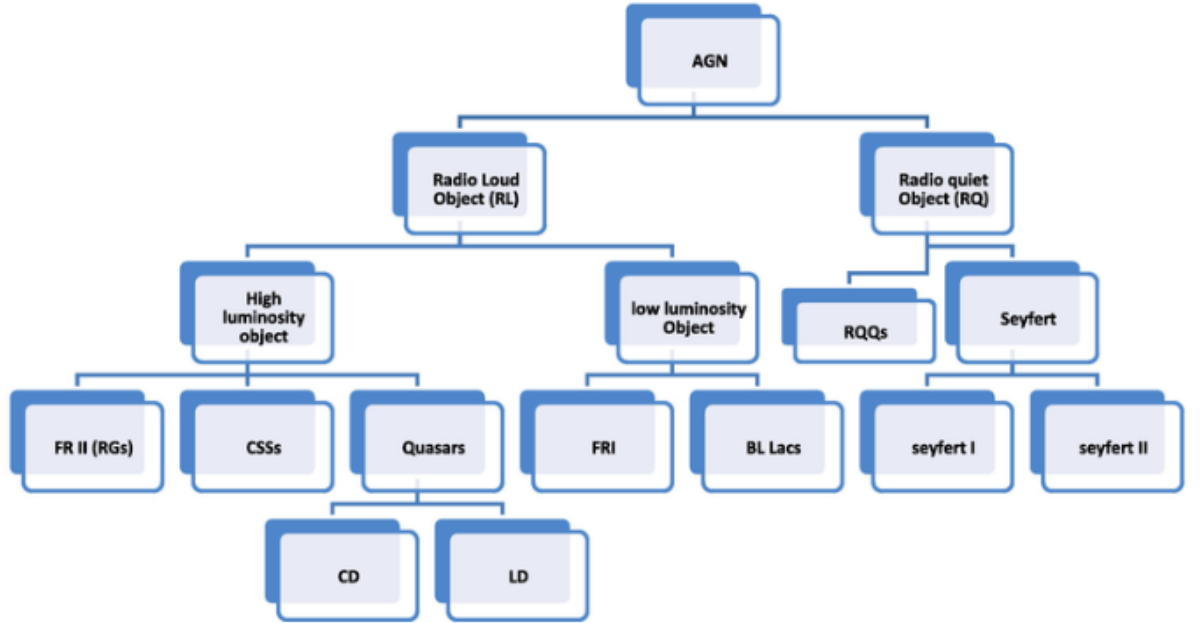


Figure 1.2: Classification of AGN

Active galactic nuclei are classified to Radio loud and Radio quiet galaxies based on the ratio of luminosity at 5 GHz to optical luminosity at B-band $L_{5GHz}/LB = 10$ is the demarcating value.

1.1.2 Radio Quiet AGNs

Radio quiet galaxies are galaxies having lower radio luminosity than optical luminosity. They are further classified based on ratio of optical luminosity to uv luminosity into Seyfert type 1 and Seyfert type 2 galaxies.

1.1.3 Radio Loud AGNs

Radio loud galaxies are galaxies having greater radio luminosity than optical luminosity. Based on width of emission lines in their spectra, they are classified into BLRG (Broad Line Radio Galaxy) and NLRG (Narrow Line Radio Galaxy), radio quasars and BL-Lacs.

1.1.4 BL Lac or BL Lacertae

BL Lac is an active galaxy with a bright and rapidly variable non-thermal emission that is dominated by a relativistic jet pointed towards the observer. They have strong nuclear continuum, high polarization and rapid flux. BL-Lac with flat spectrum Radio Quasars are collectively considered as Blazars due to their similarities such as high and variable polarization, rapid variability and superluminal motion

1.1.5 Quasars

Quasars are extremely luminous, distant objects that emit intense radiation across a wide range of wavelengths. They were first identified in the 1960s, and were initially thought to be stars. It was later realized that quasars are actually powered by supermassive black holes at the centers of galaxies, similar to other types of AGN. Quasars are classified based on the properties of their spectra, with Type 1 quasars having broad emission lines and Type 2 quasars having only narrow lines.

1.2 Blazars

Blazars are galaxies having strong varying radiations and has no strong emission or absorption lines. Due to which finding

redshift of these galaxies is very difficult. The optical luminosity of some BL Lac varies by several magnitudes. All Blazars are radio sources and also sometime show violent variability of highly energetic gamma radiation. At peak emission, observing a blazar feels a little like looking down the barrel of the most powerful gun in the universe. In some cases, the material exiting a blazar appears to be moving faster than the speed of light, but this is an illusion caused by the geometry of a high-speed source that is not oriented exactly head-on with the observer.

Blazars are classified according to their width of optical emission lines. FSRQs (Flat Spectrum Radio Quasar) have broad width emission lines which are greater than 5 Angstrom whereas BL Lac are the ones with narrow width emission line which are smaller than 5 Angstrom

1.2.1 Identifying a Blazar

The first blazar was initially mistaken as an unusual star. In 1929, Cuno Hoffmeister, a German astronomer, listed 354 objects in his catalogue that he believed to be variable stars. One of these objects was named BL Lacertae or BL Lac, based on the constellation it was located in, Lacerta. However, by the late 1960s and 1970s, astronomers noticed that BL Lac varied in brightness in an unpredictable manner and emitted a significant amount of radio waves, which was not typical for stars. Further research indicated that BL Lac was too distant to be a star in the Milky Way galaxy and its behavior was more similar to that of quasars than variable stars.

By the late 1960s and 1970s, astronomers began to notice that BL Lac get brighter and fainter, but not in a regular, predictable way, and it also seemed to emit a lot of light in the radio wave regime, which was unusual for stars. Further studies showed that BL Lac was too far away to be a star in the Milky Way galaxy.

And in some ways, its behaviour looked more like another mysterious object astronomers were finding — called quasars — than it did variable stars.

Eventually, astronomers found that BL Lac was in fact a bright object in the centre of a distant galaxy and began to find other objects that shared BL Lac’s strange properties, calling them “BL Lac objects”



Figure 1.3: CentaurusA

1.3 Radiative Process in Blazar

The spectral energy distribution of a Blazar features two peaks - the low energy peak resulting from synchrotron radiation caused by relativistic electrons losing energy in magnetic fields, and the high energy peak arising from the inverse¹⁴. Compton scattering of soft target photons by the same group of electrons.

1.4 Unification Hypothesis

The idea that all AGN types belong to the same population stems from their shared observational similarities, and the Orientation Hypothesis; provides a successful explanation for this unification. It suggests that the observed variations between AGN types are a result of differing observational perspectives. The Seyfert 1 and Seyfert 2 galaxies are grouped together as a result of broad emission lines being detected in the polarized spectra of Seyfert 2. Thus, the Orientation theory posits that the regions producing the broad emission lines are located close to the central nucleus and obscured by the dark torus. In contrast, the region responsible for producing the narrow emission lines is far from the central engine and not covered by the dusty torus.

The overall depiction of AGNs features dual jets, with quasar jets oriented closer to the line of sight than those from FR-2 radio quasars, resulting in a relativistic beaming effect. Conversely, Blazars are line-of-sight aligned versions of FR-1 radio quasars.

1.5 Jets Of Active Galactic Nucleus

In the event of a blazar, where the jets are aimed towards the line of sight, it has been noted that the jet components move at speeds surpassing the speed of light. This occurrence is referred to as superluminal motion. The correlation between the apparent velocity of the radiation blob, and angle of the radiation blob relative to the line of sight, is determined.

$$\beta_a = \frac{\beta \sin \theta}{1 - \beta \cos \theta} \quad (1.1)$$

β_a and β are the dimensionless velocities and subscript a representing apparent velocities. The dependence of β_a on the view-

ing angle θ has been plotted in the following figure.

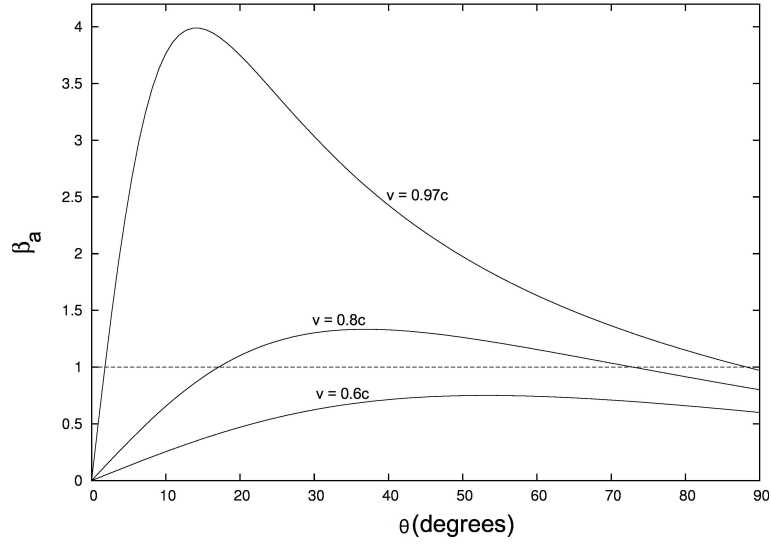


Figure 1.4: Apparent velocity of a relativistically moving blob (in units of c) with respect to the viewing angle of the observer. Each curve represents blob with actual velocity $0.6c$, $0.8c$ and $0.97c$.

Chapter 2

RADIATIVE PROCESS AND BLAZAR SED

2.1 Synchrotron

Cyclotron radiation is a type of non-thermal emission that is produced when charged particles, such as electrons or protons, move in a magnetic field. When charged particles move in a magnetic field, they experience a force perpendicular to both their velocity and the magnetic field. This force causes the charged particles to spiral along the magnetic field lines, emitting radiation as they accelerate. Cyclotron radiation is typically observed at radio wavelengths and is characterized by its polarization, which is perpendicular to the magnetic field. The intensity of cyclotron radiation depends on the strength of the magnetic field and the energy of the charged particles emitting the radiation.

Synchrotron radiation is produced when charged particles are accelerated in a curved path by a magnetic field that changes direction. The curvature of the path causes the charged particles to emit radiation tangentially to their motion, resulting in a broad spectrum of radiation that extends from radio waves to gamma rays. The frequency of the radiation depends on the energy of the charged particle, the strength of the magnetic field,

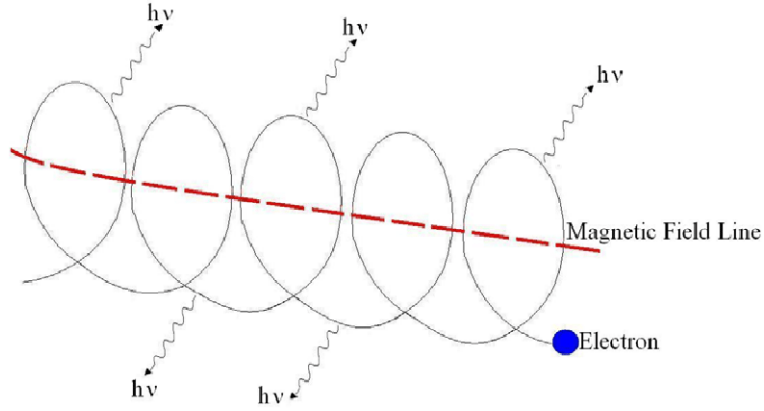


Figure 2.1: Credits: Çelik, Celik, Murat Technology, Ph. (2021). Experimental and computational studies of electric thruster plasma radiation emission.

and the curvature of the particle's path.

The main difference between cyclotron and synchrotron radiation is that cyclotron radiation occurs when a charged particle moves in a uniform magnetic field, while synchrotron radiation occurs when a charged particle moves in a magnetic field that varies in strength and direction. In the case of high-velocity electrons moving in a magnetic field, the radiation they emit is called synchrotron radiation rather than cyclotron radiation. This is because the electrons' motion is not restricted to a circular path, as in the case of cyclotron radiation. Instead, they move along a curved trajectory, emitting radiation at each turn. The emitted radiation has a characteristic angular power pattern that differs from that of cyclotron radiation. Synchrotron is a relativistic cyclotron radiation.

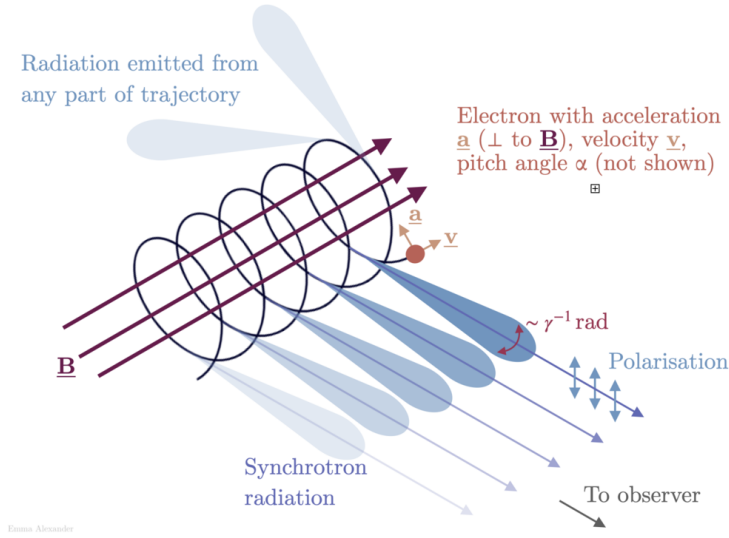


Figure 2.2: Synchrotron emission

2.2 Relativistic Beaming effect:

An electron accelerating in magnetic field radiates electro magnetic field radiation , and shows different effect

2.2.1 Length Contraction

Length contraction is a phenomenon predicted by Einstein's theory of special relativity, which states that objects appear shorter in the direction of motion as measured by a stationary observer. In the case of synchrotron radiation, charged particles such as electrons are accelerated to relativistic speeds in a circular path by a magnetic field. As these particles move through the magnetic field, they experience a Lorentz contraction in the direction of their motion, which compresses the magnetic field lines.

The compression of the magnetic field lines results in an increase in the strength of the radiation emitted by the charged particles. This enhanced radiation is observed in synchrotron sources as a result of the relativistic motion of the particles.

2.2.2 Doppler Shift

The Doppler effect is a phenomenon that occurs when there is relative motion between the emitter and the observer of radiation. In the case of synchrotron radiation, the charged particles emitting the radiation are often moving at relativistic speeds, causing the radiation to be shifted to higher or lower frequencies depending on the direction of motion.

When a charged particle emits synchrotron radiation while moving towards an observer, the

$$\Delta t^A = 2/(\gamma\omega_g \sin(\alpha))(1/(2\gamma^2))$$

the radiation is compressed in frequency, resulting in a higher frequency than if the particle were stationary. Conversely, if the charged particle emits radiation while moving away from the observer, the radiation is stretched in frequency, resulting in a lower frequency than if the particle were stationary. This shift in frequency is known as the Doppler shift.

2.3 Synchrotron Spectrum

The emitted radiation of cyclotron frequency and the angular frequency of gyration are identical, appearing as simple sinusoidal waves. However, synchrotron radiation exhibits a beaming effect, causing the observed radiation frequency to be much lower than the angular frequency of gyration. The angular frequency of gyration can be calculated using the formula $\omega_g = qB/mc$, where q is the electron charge, B is the magnetic field, γ is the relativistic factor, and m is the mass of an electron. When an electron moves with relativistic velocities, the emitted radiation is visible for only a brief moment, appearing between two points, p_1 and p_2 . The time an electron takes to travel between these two points is given by $t_2 - t_1 = 2/(\gamma\omega_g \sin(\alpha))$ whereas the

time for radiation to be observed is not the same.

$$t_2^A - t_1^A = 2/(\gamma\omega_g \sin(\alpha))(1 - u/c)$$

$$\Delta t^A = 2/(\gamma\omega_g \sin(\alpha))(1/(2\gamma^2))$$

Now we can derive the critical frequency

$$\omega_c 3/2\gamma^3 \omega_g \sin(\alpha)$$

2.4 Synchrotron Self Compton

Synchrotron self-Compton is a phenomenon where synchrotron photons are scattered by the same population of relativistic electrons that originally produced them. The process of inverse Compton scattering is responsible for this phenomenon, where high-energy electrons scatter photons, increasing their energy. This phenomenon is the opposite of Compton scattering, where high-energy photons knock off an electron, leading to a lower energy photon and an ionized electron.

In the case of blazar jets, the increased energy of these photons pushes them into the X-ray regime. The Synchrotron self-Compton model is derived in a very similar way to Synchrotron radiation, The Synchrotron self-Compton model predicts a double-peaked spectral energy distribution for blazars, where the first peak corresponds to the synchrotron radiation, and the second peak corresponds to the inverse Compton scattering of the synchrotron photons by the same population of relativistic electrons. This phenomenon provides crucial insights into the physical processes occurring in blazar jets and the nature of the emission mechanisms.

2.5 Blazar Spectral Energy Distribution

A Blazar's spectral energy distribution has two bumps one is the low energy bump which comes from synchrotron radiation from

relativistic electrons losing its energy in the magnetic field and another high energy bump which comes from inverse Compton scattering of soft target photons by the same electron distribution.

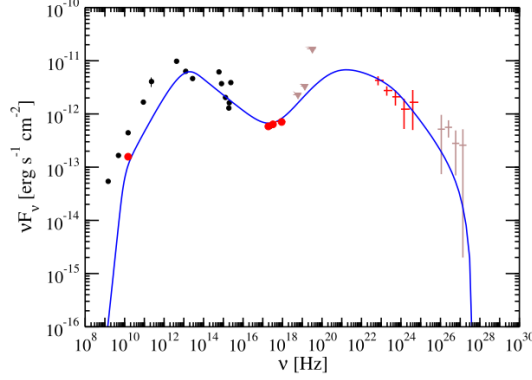


Figure 2.3: Multiwavelength radio-galaxy SEDs. SED data of M87

2.6 Blazar Jet Models

The spectrum of the emission from a blazar comes due to the broken power law distribution of electrons. So, we define broken power law distribution of electrons as following.

$$N(\gamma)d\gamma = \begin{cases} K\gamma^{-p}d\gamma & \text{for } \gamma_{\min} < \gamma < \gamma_b \\ K\gamma_b^{q-p}\gamma^{-q}d\gamma & \text{for } \gamma_b < \gamma < \gamma_{\max} \end{cases} \text{ cm}^{-3} \quad (2.1)$$

where K is the normalization factor, $\gamma = \frac{E}{m_e c^2}$ is dimensionless energy m_e is the mass of an electron, c being the speed of light, p and q are low and high spectral indices respectively. non-thermal emission from the blazar to originate from a spherical region of radius R , moving down the jet with bulk Lorentz factor Γ at an angle θ with respect to the line of sight of the observer.

2.6.1 Synchrotron Emissivity

The Synchrotron emissivity of isotropic electron distribution in magnetic field B is given by

$$j_{\text{syn}}(\nu) = \frac{1}{4\pi} \int_{\gamma_{\text{min}}}^{\gamma_{\text{max}}} P_{\text{syn}}(\gamma, \nu) N(\gamma) d\gamma \quad (2.2)$$

erg cm⁻³ s⁻¹ Hz⁻¹ Sr⁻¹

here $P_{\text{syn}}(\gamma, \nu)$ is pitch angle averaged single particle emissivity, which is given by

$$P_{\text{syn}}(\gamma, \nu) = \frac{\sqrt{3}\pi e^3 B}{4m_e c^2} F\left(\frac{\nu}{\nu_c}\right) \quad (2.3)$$

where ν_c is equal to

$$\nu_c = \frac{3\gamma^2 e B}{16m_e c} \quad (2.4)$$

here F is the synchrotron power function and is given by

$$F(x) = 1.8x^{(\frac{1}{3})}e^{-x} \quad (2.5)$$

Chapter 3

Research Methodology

3.1 Objective of Research

The spectral energy distribution of blazars extends from radio-to-gamma-rays. The lower energy extending from radio to x-ray are well understood to be by synchrotron emission processes. An approximate analytical solution to the synchrotron emissivity is calculated. From the observed spectral shape and its features and also using approximate synchrotron emissivity, we estimate/constrain the physical parameters of the source.

3.2 Approximate Analytical Solution

The analytical solution to synchrotron emissivity is given by, The Synchrotron emissivity of isotropic electron distribution in magnetic field B is given by

$$j_{\text{syn}}(\nu) = \frac{1}{4\pi} \int_{\gamma_{\text{min}}}^{\gamma_{\text{max}}} P_{\text{syn}}(\gamma, \nu) N(\gamma) d\gamma \quad (3.1)$$

$\text{erg cm}^{-3} \text{ s}^{-1} \text{ Hz}^{-1} \text{ Sr}^{-1}$

here $P_{\text{syn}}(\gamma, \nu)$ is pitch angle averaged single particle emissiv-

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$$\nu_c = \frac{3\gamma^2 e B}{16m_e c} \quad (3.3)$$

here F is the synchrotron power function and is given by

$$F(x) = 1.8x^{(\frac{1}{3})}e^{-x} \quad (3.4)$$

Equation 3.1 can be solved numerically, but it does not give any information of involved parameters like magnetic field, energy of particle, etc. Figure 3.1 shows the power function given by equa-

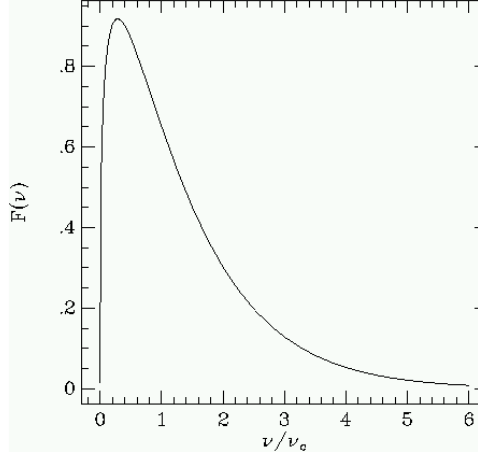


Figure 3.1: Synchrotron Power function

tion 3.4. This power function has the form of Bessel function, which is difficult to solve analytically. So we approximate this function to a delta function and solve for approximate analytical solution for emissivity.

$$P_{syn} = \frac{4}{3}\beta^2\gamma^2 c\sigma_T U_B \phi_\nu(\gamma) \quad (3.5)$$

$\beta = \frac{u}{c}$ and σ_T is the thomsons cross section. $\phi_\nu(\gamma)$ is the spectral function which satisfies the equation

$$\int_0^\infty \phi_\nu(\gamma) d\gamma = 1 \quad (3.6)$$

As the synchrotron power function $F(x)$ peaks at $x \approx 0.29$ This allows us to approximate $\phi_\nu(\gamma)$ to a δ function.

$$\phi_\nu(\gamma) \rightarrow \delta(\nu - \gamma^2 \nu_L) \quad (3.7)$$

We can use δ function property to solve the equation. ν_L is the larmor frequency

$$\equiv \frac{eB}{2\pi m_e c}$$

$$P_{syn} = \frac{4}{3} \beta^2 \gamma^2 c \sigma_T U_B \delta(\nu - \gamma^2 \nu_L) \quad (3.8)$$

delta function property

$$\delta(\phi_\nu(\gamma)) = \sum_i \frac{\delta[\gamma - \gamma_i]}{\frac{d\phi_\nu(\gamma)}{d\gamma}} \quad \gamma = \gamma_i \quad (3.9)$$

γ_i are the roots of the spectral function. Substituting $\delta(\phi_\nu(\gamma))$ in equation 2.9 . We get Hence equation 2.2 implies

$$j_{syn} \approx \frac{\sigma_T c B^2}{48\pi^2} \nu_L^{-\frac{3}{2}} N(\sqrt{\frac{\nu}{\nu_L}}) \nu^{\frac{1}{2}} \quad (3.10)$$

3.2.1 Observed Synchrotron Flux

The observed flux for synchrotron emission processes can be determined by using the synchrotron emissivity. Doppler and cosmological effects are considered the observed flux at ν_{obs} in the direction Ω_{obs} is given by

$$F_{obs}(\nu_{obs}) = \frac{\delta_D^3 (1+z)}{d_L^2} V j_{rad}(\frac{1+z}{\delta_D} \nu_{obs}, \mu, \phi_{obs}) \quad (3.11)$$

Substituting $V = \frac{4}{3}\pi R^3$ and j_{rad} from equation 8 into this observed flux equation and solving we get.

$$F_{\text{obs}}^{\text{syn}}(\nu_{\text{obs}}) \approx \begin{cases} S(z, p) \delta_D^{\frac{p+5}{2}} B^{\frac{p+1}{2}} R^3 K \nu_{\text{obs}}^{-\left(\frac{p-1}{2}\right)} & \text{for } \nu_{\text{obs}} \ll \delta_D \gamma_b^2 \nu_L / (1+z) \\ S(z, q) \delta_D^{\frac{q+5}{2}} B^{\frac{q+1}{2}} R^3 K \gamma_b^{q-p} \nu_{\text{obs}}^{-\left(\frac{q-1}{2}\right)} & \text{for } \nu_{\text{obs}} \gg \delta_D \gamma_b^2 \nu_L / (1+z) \end{cases} \quad (3.12)$$

3.2.2 Observed Synchrotron Self Compton Flux

Using the same equation we derive the observed flux for SSC.

$$F_{\text{obs}}^{\text{SSC}}(\nu_{\text{obs}}) \approx \begin{cases} C(z, p) \delta_D^{\frac{p+5}{2}} B^{\frac{p+1}{2}} R^4 K^2 \nu_{\text{obs}}^{-\left(\frac{p-1}{2}\right)} \log\left(\frac{\gamma_b}{\gamma_{\text{min}}}\right) & \text{for } \nu_{\text{obs}} \ll \delta_D \gamma_b^4 \nu_L / (1+z) \\ C(z, q) \delta_D^{\frac{q+5}{2}} B^{\frac{q+1}{2}} R^4 K^2 \gamma_b^{2(q-p)} \nu_{\text{obs}}^{-\left(\frac{q-1}{2}\right)} \log\left(\frac{\gamma_{\text{max}}}{\gamma_b}\right) & \text{for } \nu_{\text{obs}} \gg \delta_D \gamma_b^4 \nu_L / (1+z) \end{cases} \quad (3.13)$$

3.2.3 Data Extraction

Radio, Optical, X-ray spectral energy distribution data of few bright blazars were collected using the tool XY scan.

3.2.4 SED Fitting

The approximate analytical solution to synchrotron flux coded using FORTRAN(readymade code) is used along with the extracted observed data for the SED fitting.

Chapter 4

Results and Discussion

In this project we have derived an analytical model by which we constrained few source parameters. The numerical model developed and used by Sahanayathan and Godambe, 2012 is also used to study. The analytical model gives us a sense of idea about how different parameters affect a blazar's broadband spectrum. The analytical solutions of (3.12) and (3.13) shows us the dependency about how one parameter to the spectral shape.

4.1 Parameter dependence

4.1.1 Radius R

The relation between radius of the emitting region (R) and the observed synchrotron flux or F_{obs}^{syn} is such that, by increasing the radius by a factor of 2, the observed synchrotron flux increases by a factor of 2^3 i.e. 8 and for the synchrotron case and by a factor of 2^4 i.e. 16 for the Synchrotron Self Compton case.

4.1.2 Normalization Factor K

When the normalization factor increased by a factor of 2 the F_{obs}^{syn} would increase by a factor of 2 but in the case of SSC F_{obs}^{ssc} would increase by 4. This would shift the whole broadband spectrum up.

4.1.3 Spectral Indices p and q

Rather than increasing or decreasing the values of our spectral indices we would try to understand how the values of both affect the broadband spectrum.

Increase in p increases the angle between the left hand positive slope wrt to the right hand negative slope. When p is equal to q it becomes a straight line which means it's a power law distribution rather than broken power power law. Further increase in p would lead to formation of v-type plot which is physically impossible.

We get a better understanding of how spectral indices shape the graph by applying log to the equation.

$$\log(F_{syn}^{obs}) \propto (\frac{p+5}{2})\log\delta_D + (\frac{p+1}{2})\log B + 3\log R + \log K + (\frac{1-p}{2})\nu_{obs} \quad (4.1)$$

and

$$\log(F_{syn}^{obs}) \propto (\frac{q+5}{2})\log\delta_D + (\frac{q+1}{2})\log B + 3\log R + \log K + (q-p)\log\gamma_b + (\frac{1-q}{2})\log\nu_{obs} \quad (4.2)$$

from these equations we understand that p and q are the slopes of the broadband spectrum.

4.1.4 Magnetic Field B

The relation between observed flux and magnetic field depends upon the value of the spectral indices. The effect of increase in magnetic field. From (4.1) and (4.2) we can conclude that by doubling the magnetic field for the relation $\log(F_{syn}^{obs}) \propto \frac{p+1}{2}\log B$ and $\log(F_{syn}^{obs}) \propto (\frac{q+1}{2})\log B$. The whole broadband spectrum would shift upward by $(\frac{p+1}{2}) \times \log 2$ and $(\frac{q+1}{2}) \times \log 2$.

4.1.5 Doppler Factor δ_D

The relation between observed flux and doppler factor also depends upon the value of the spectral indices. From (4.1) and (4.2) we can conclude that by doubling the doppler factor for the relation $\log(F_{syn}^{obs}) \propto (\frac{p+5}{2})\log\delta_D$ and $\log(F_{syn}^{obs}) \propto (\frac{q+5}{2})\log\delta_D$. The whole broadband spectrum would shift upward by $(\frac{p+5}{2}) \times \log 2$ and $(\frac{q+5}{2}) \times \log 2$.

4.2 Numerical Model Results

We tried to fit the numerical model on Mkn 421 flux data points which was developed on Fortran. With the help of gnuplot the fitting was done on Linux Ubuntu P.C. The Numerical model helped us to constrained few parameters. The synchrotron and synchrotron self compton model curves plotted on Mkn 421 are shown in Fig. below. The parameter constraints are given in table.

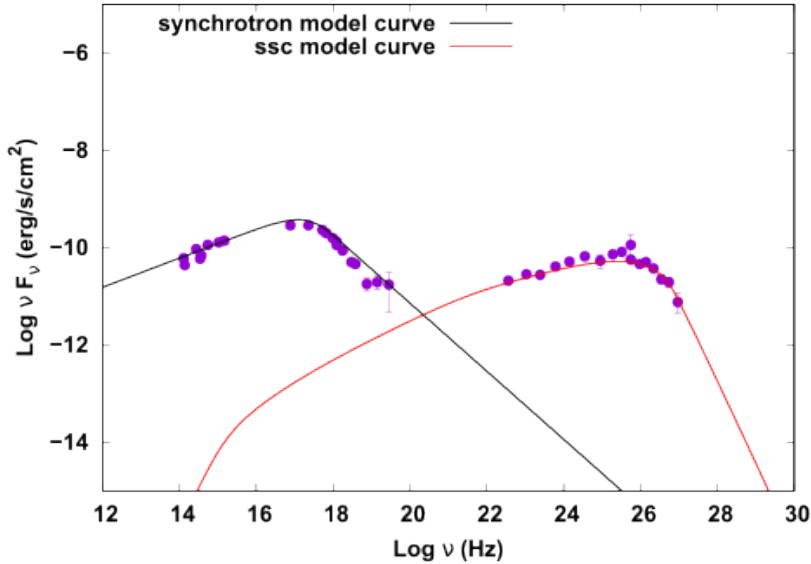


Figure 4.1: Spectral energy distribution of Mkn 421 with Synchrotron and Synchrotron Self Compton model curve

Parameters	Values
p (lower spectral index)	0.70
q (higher spectral index)	1.70
Magnetic Field	$0.065G$
Doppler Factor δ_D	71.962
Particle Normalization	3.5×10^4

Table 4.1: Parameters and their values for Mkn 421

Chapter 5

Conclusion

In the present work, we develop a statistical fitting procedure of the broadband spectrum of blazars, considering Synchrotron and SSC emission mechanisms.

We fitted the numerical solutions to constrain parameters to understand relationship between parameters and broadband spectral energy distribution shape. We found the values of those parameters responsible for SED of Mkn421 2011 emission.

5.1 Future scope

The future scope of studying the non thermal emission of blazars is vast. Here are some possible areas of research that could be explored such as understanding the mechanisms of particle acceleration. The exact mechanism by which particles are accelerated in blazars is still not fully understood. Studying the non-thermal emission of blazars can provide clues about the physical processes that are at work in these objects. Probing the structure of blazar jets. Blazar jets are believed to be the source of the non-thermal emission in these objects. Studying the non-thermal emission can provide information about the structure and dynamics of these jets.

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